Growing Cents: Technical Sequence Diagrams

This document outlines the core operational flows within the Growing Cents ecosystem, mapping interactions between the **High Command (Admin)**, **Institutional Nodes (Clubs)**, and **Strategic Intelligence (Assessment Engine)**.

1. Authentication & Security Protocol (Login)

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mermaid

sequenceDiagram

    participant User

    participant AuthController

    participant 2FAService

    participant AuditLog

    participant Dashboard
```

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User->>AuthController: Submit Credentials (email/pass)
AuthController->>AuthController: Validate Credentials
alt Credentials Valid
    AuthController->>2FAService: Trigger OTP Token
    2FAService-->>User: Dispatch SMS/Email Code
    User->>AuthController: Submit Verification Code
    AuthController->>2FAService: Verify Token
    alt Token Valid
        AuthController->>AuditLog: Store 'login_success'
        AuthController->>Dashboard: Redirect to Console
        Dashboard-->>User: Render Dashboard
    else Token Invalid
        AuthController->>AuditLog: Store '2fa_failure'
        AuthController-->>User: Return Error (422)
    end
else Credentials Invalid
    AuthController-->>User: Return Error (401)
end
```

2. Institutional Audit Workflow (Assessment Submit)

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sequenceDiagram

    participant Club Admin

    participant AssessmentController

    participant MaturityEngine

    participant Database

    participant IntelligenceHub



    Club Admin->>AssessmentController: Initialize Audit (GET /create)

    AssessmentController-->>Club Admin: Render Diagnostic Framework

    Club Admin->>AssessmentController: Submit Responses (POST /store)

    AssessmentController->>MaturityEngine: Process Raw Data

    Maturity Engine->>Maturity Engine: Calculate Section Scores

    Maturity Engine->>Maturity Engine: Derive Global Maturity Score (%)

    AssessmentController->>Database: Persist 'assessment_record' (status

    AssessmentController->>IntelligenceHub: Generate Strategic Recommenda
```

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    IntelligenceHub-->>AssessmentController: Return Optimization Protoco

    AssessmentController-->>Club Admin: Redirect to Success Roadmap (200
```

3. Strategic Advisory Generation (Report & PDF)

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sequenceDiagram

    participant Requester

    participant AssessmentController

    participant PdfEngine

    participant Storage

    participant CDN


    Requester->>AssessmentController: Request Export (GET /export)

    AssessmentController->>AssessmentController: Verify Permissions

    AssessmentController->>AssessmentController: Fetch Assessment + Reco
```

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    AssessmentController->>PdfEngine: Inject Data into Template

    PdfEngine->>PdfEngine: Render 'Elite' PDF Layout

    PdfEngine->>Storage: Temporarily Cache Asset

    AssessmentController->>CDN: stream_file()

    CDN-->>Requester: Download File (application/pdf)
```

4. Club Institutional Onboarding (Registration)

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sequenceDiagram

    participant User

    participant ClubController

    participant Database


    User->>ClubController: Register Club (POST /register)

    ClubController->>ClubController: Validate Institution Data
```

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    ClubController->>Database: Create Club Record

    ClubController->>Database: Assign User as L4 (President)

    ClubController-->>User: Redirect to Onboarding Wizard
```

5. Multi-Club Governance Protocol

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sequenceDiagram

    participant Admin

    participant RegionalController

    participant Aggregator


    Admin->>RegionalController: Select Region

    RegionalController->>Aggregator: Fetch Multi-Club Metrics

    Aggregator->>Database: Query 'maturity_levels' by Region

    Aggregator-->>RegionalController: Return Aggregated Yield

    RegionalController-->>Admin: Render Global Cockpit
```